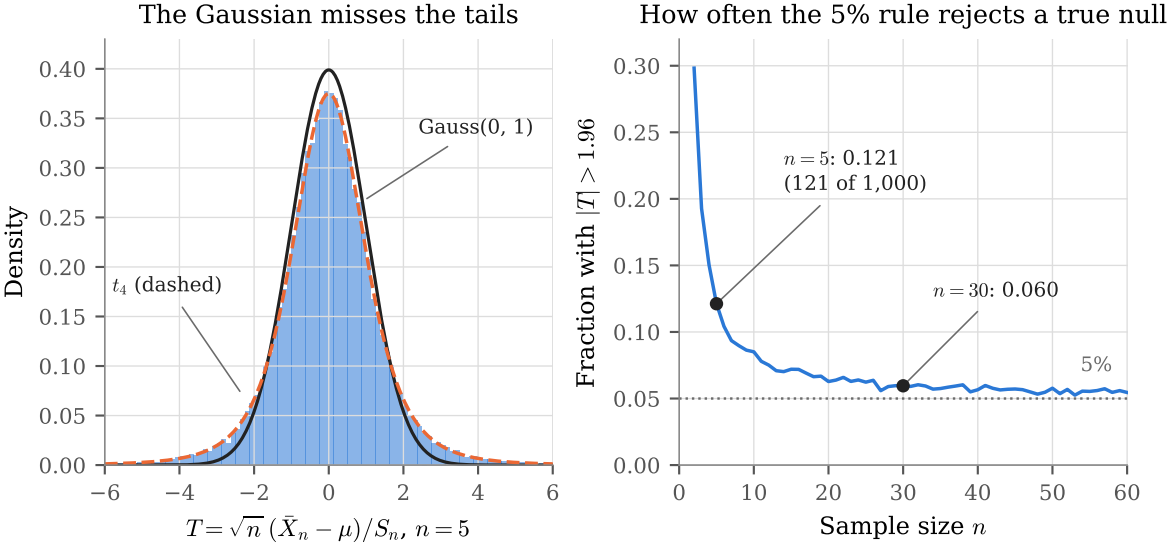
The Gaussian misses the tails
Density
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Gauss(0, 1)
t_4 (dashed)
−6
−4
−2
0
2
4
6
$T = \sqrt{n}\,(\bar{X}_n - \mu)/S_n$, $n = 5$
How often the 5% rule rejects a true null
Fraction with $|T| > 1.96$
0.30
0.25
0.20
0.15
0.10
0.05
0.00
$n = 5$: 0.121
(121 of 1,000)
$n = 30$: 0.060
5%
0
10
20
30
40
50
60
Sample size n